

Impact statement

Experimental investigation on the vortex-induced vibration of a rectangular 4:1 cross-section continuous girder bridge in natural wind field

Bo Wang^a, Fuyou Xu^{b,*}, Mingjie Zhang^b

^a Mechanical Engineering College, Xi'an Shiyou University, Xi'an 710065, China

^b School of Civil Engineering, Dalian University of Technology, Dalian 116024, China

ARTICLE INFO

Keywords:

Natural wind field

Full-bridge aeroelastic model

Rectangular 4:1 cross-section

Vortex-induced vibration

ABSTRACT

Significant vortex-induced vibrations (VIV) occur in certain long-span continuous girder bridges exposed to frequent wind speeds. Investigating the VIV response of these bridges under natural wind conditions is crucial for assessing their wind resistance performance. The VIV characteristics of a rectangular 4:1 cross-section continuous girder bridge (R4:1-CGB) were studied in a representative natural wind field. The wind conditions at the test site were thoroughly analyzed. The influence of natural wind parameters on VIV was evaluated using synchronized measurements of wind data and vibration signals. The results indicated that under natural wind conditions, the VIV mode is predominantly characterized by a single mode, with a distinct dominant frequency, although the amplitude fluctuates with wind speed. The observed wind speed and VIV amplitude generally fall within the VIV range identified in wind tunnel section model tests. The amplitude increases with wind speed but remains notably lower than the results obtained from wind tunnel tests. The maximum wind yaw angle for VIV is 50°, and the VIV amplitude increases as the wind direction becomes more perpendicular to the bridge axis. At similar wind speeds, the VIV amplitude increases as turbulence intensity decreases and wind angles of attack increase.

1. Introduction

Wind tunnel tests and field measurements are essential methods for studying vortex-induced vibrations (VIV) in long-span bridges, each offering distinct advantages and complement one another. Wind tunnel tests primarily employ scaled models to investigate the aerodynamic performance of the bridge, including rigid sectional models [1–3] and full-bridge aeroelastic model tests [4,5]. The scale effect in a wind tunnel makes it difficult to replicate actual wind conditions, structural dynamics, and high Reynolds numbers, among other factors. It also limits the accurate simulation of structural details, such as girder guide angles and railings, potentially leading to inaccuracies in test results [6,7]. In contrast, field measurements can overcome the limitations of wind tunnel tests by providing real-world data on bridge VIV. They can capture the actual VIV response of the bridge and validate the results from wind tunnel tests. However, field measurements can be costly and influenced by uncontrollable wind conditions [8,9].

Due to the aerodynamic complexity and geometric variability of real bridge decks, the 4:1 rectangular cylinder is often considered a simplified bridge deck because its VIV response and excitation mechanism are

similar to those of bluff bridge decks [10–12]. Wang et al. [13] experimentally investigated the VIV characteristics of a 4:1 rectangular cylinder under skew wind using a sectional model. The results suggest that VIV is most severe when the wind is perpendicular to the model's axis. However, it remains unclear whether this conclusion holds in a real-world wind field environment. Xu et al. [9] proposed a new experimental approach to study the wind-induced vibrations of a large-scale full-bridge aeroelastic model in a natural wind field. The scale ratio of this new approach is significantly larger than that of wind tunnel tests. Additionally, it is more cost-effective than field measurements and allows for structural parameter adjustments.

This paper investigates the VIV of a rectangular 4:1 cross-section continuous girder bridge (R4:1-CGB) in the natural wind field using a full-bridge aeroelastic model. After long-term monitoring, multiple VIV response datasets of the model were obtained and compared with wind tunnel test results. The primary objectives of this study include: (1) assessing the feasibility of using a full-bridge aeroelastic model in a natural wind field for VIV research; (2) analyzing the differences between the VIV responses of a 4:1 rectangular cylinder in a natural wind field and in a wind tunnel.

* Corresponding author.

E-mail address: fuyouxu@dlut.edu.cn (F. Xu).

<https://doi.org/10.1016/j.istruc.2025.110278>

Received 11 June 2025; Received in revised form 2 September 2025; Accepted 21 September 2025

Available online 23 September 2025

2352-0124/© 2025 Institution of Structural Engineers. Published by Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.



Fig. 1. The location of the experimental site.

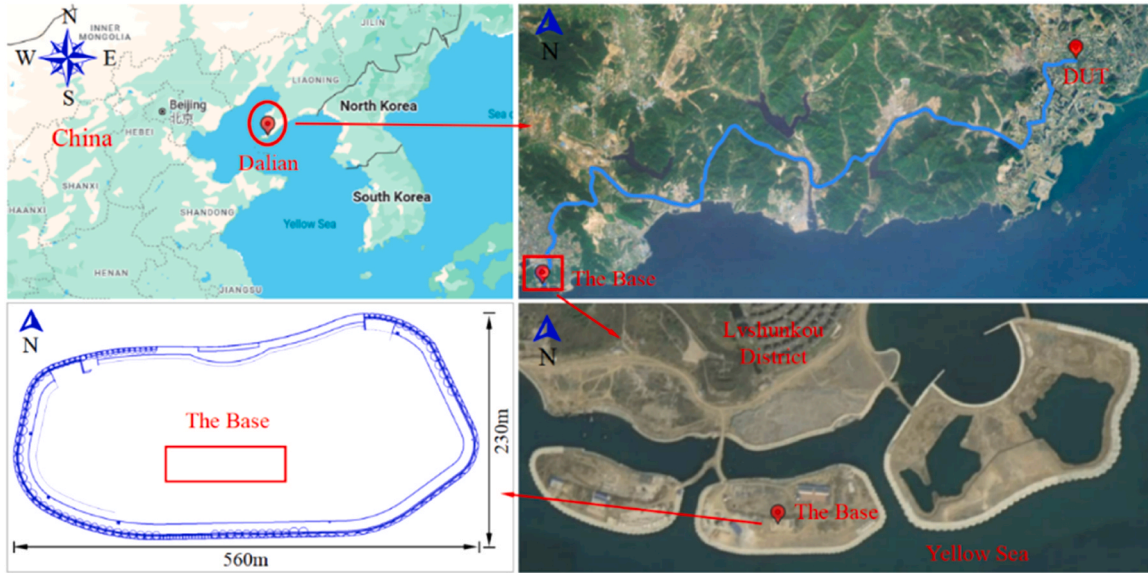


Fig. 2. A picture of full-bridge aeroelastic model.

2. Test overview

2.1. Experimental Base

The experimental site is located on an island in Dalian City, Liaoning Province, China, approximately 36 km from Dalian University of Technology. The location of the experimental site is shown in Fig. 1. Details of the experimental site were presented in another publication [9] and are not included in this study.

2.2. Full-bridge aeroelastic model and parameter identification

The height and width of the cross section of the rectangular cylinder are 30 cm and 120 cm, respectively. The full-bridge aeroelastic model has a span configuration of 15 m + 20 m + 15 m. It features aluminum core beams to provide the stiffness, and high-density foam plates and waterproof high-strength wallpaper to replicate the aerodynamic shape. The model consists of an aluminum square tube frame arranged in two rows, each with 8 beam segments. Each row has two 4.35 m non-standard sections at the ends, with the remaining sections being 6 m in length. The beam segments are connected by welding and further reinforced with localized reinforcement plates and bolts. Multi-channel

transverse connecting rods link the two rows of beam segments to ensure structural stiffness. The mass of the model is 15 kg/m. The overall model is shown in Fig. 2.

This paper focuses on the first-order vertical VIV of the R4:1-CGB. Before conducting the tests, the first-order vertical bending frequency and damping ratio of the model were measured. The first-order vertical bending frequency was found to be 2.22 Hz, the damping ratio (ξ_0) was 0.7 %, and the Scruton number of the present model is $Sc = 4\pi M \xi_0 / (\rho BDL) = 3.1$.

2.3. Test equipment and measuring point arrangement

The Monitoring equipment includes 13 unidirectional piezoelectric acceleration sensors, each having a range of 10 g and a sensitivity of 490 mV/g. The arrangement of these sensors is shown in Fig. 3, with the red squares denoting the sensors (labeled A1 to A13). A 64-channel acquisition instrument, integrated with a signal conditioner, operates within a frequency range of 0.2–2500 Hz, powered by a 24 V power supply. It is connected to a mobile power source and a laptop for continuous monitoring. During testing, the sampling frequency of the sensors is set to 200 Hz. The vertical VIV displacement amplitude at each measurement point can be calculated from the filtered vertical

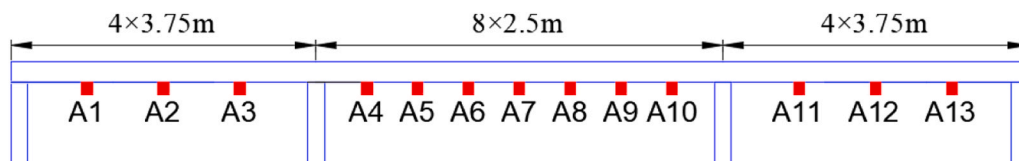


Fig. 3. Acceleration sensors layout.

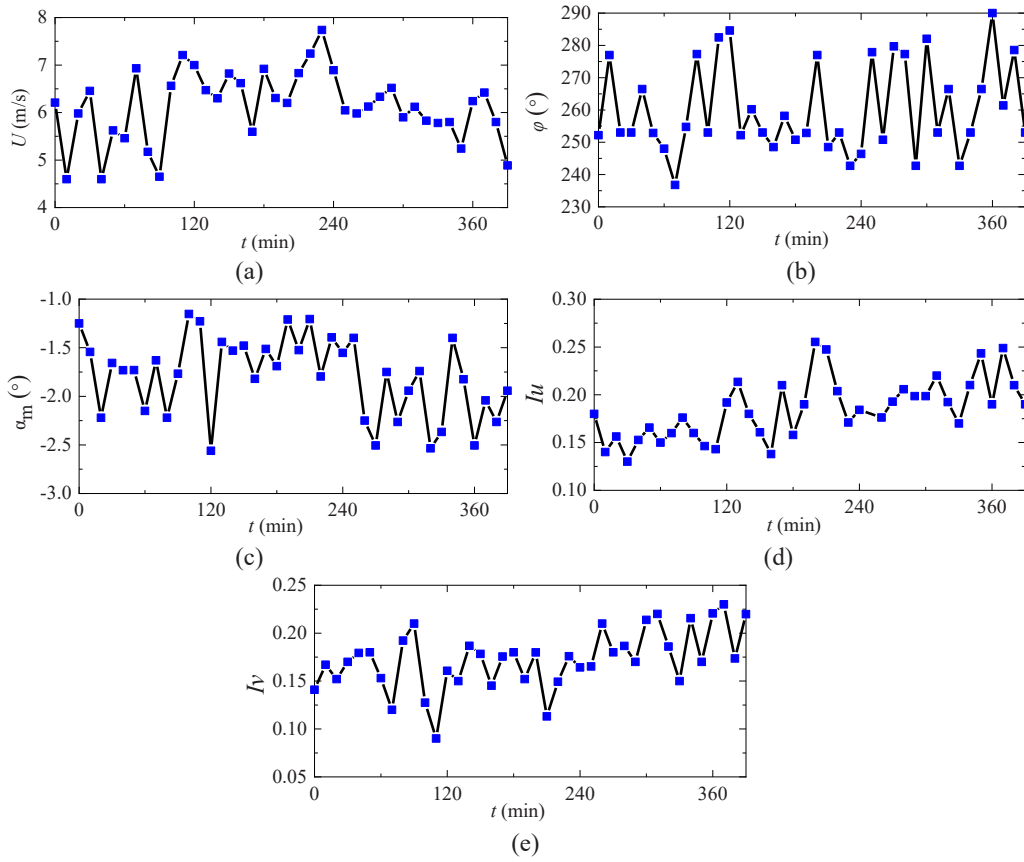


Fig. 4. Wind field data (2023.10.17): (a) U ; (b) φ ; (c) α_m ; (d) I_u ; (e) I_v .

acceleration and the dominant vibration frequency. The Equation for this calculation is as follows:

$$A = A_a / (2\pi f_v)^2 \quad (1)$$

Where A is the vertical VIV displacement amplitude, A_a is the amplitude of the filtered acceleration signal, and f_v is the monotonic frequency of the acceleration signal.

To monitor the wind field characteristics at the model bridge site during testing, a wind speed measurement system was set up at the natural wind field test base. This system helps analyze how the on-site wind environment affects the VIV of R4:1-CGB. Three three-dimensional pulsating anemometers were placed along the bridge model's axis, centrally positioned in the three-span section, aligned with the incoming wind direction. Each anemometer was placed 1 m from the model and 20 cm above it. The north direction of each anemometer was parallel to the bridge model's axis. During the test, the wind speed the wind speed associated with VIV ranges from 4.1 to 7.8 m/s, so the Reynolds number ($Re = UD/\nu$, where U and ν are the inflow freestream wind velocity and dynamic viscosity, respectively) ranges from 8.31×10^4 – 1.58×10^5 . The Reynolds number effect causes changes in the boundary layer thickness of the bridge cross-section; thus, its influence on the aerodynamic characteristics of the cross-section cannot be ignored. Most studies on bridge VIV are conducted based on scaled-down models, where the Reynolds number is relatively low, inevitably raising questions about the accuracy of the results. In contrast, the large-scale full-bridge aeroelastic model test under natural wind fields further increases the test Reynolds number, thereby enhancing the reliability of the experimental results.

3. Test results and analysis

3.1. Wind field characteristics at the site

After completing the model construction, wind field and model vibration data were collected based on weather forecasts to select dates (September 19–22, 2023, and October 14–18, 2023) when wind conditions were likely to induce VIV in the model. After removing erroneous data from the pulsating anemometers, the average wind speeds for 10-minute intervals were calculated using Eq. (2). Wind field characteristics, including the wind direction angle (φ) and the angles of attack (α_m), were recorded, with calculations based on Eq. (3). Turbulence intensity, a critical parameter influencing VIV, characterizes wind speed fluctuations and is affected by surface roughness and height above ground. It was calculated using Eq. (4). The turbulence length scales (L_u , L_v , and L_w) give the average length of turbulent eddies in the airflow, and their size determines the impact area of fluctuating components on the structure. The turbulence length scales can be calculated using Eq. (5).

$$U = \sqrt{\overline{u_x(t)^2} + \overline{u_y(t)^2}} \quad (2a)$$

$$W = \overline{u_z(t)} \quad (2b)$$

where $u_x(t)$, $u_y(t)$ and $u_z(t)$ represent the data recorded by the three-dimensional anemometer. $\overline{u_x(t)}$, $\overline{u_y(t)}$ and $\overline{u_z(t)}$ denote the average values over 10 min. U represents the horizontal average wind speed, while W denotes the vertical average wind speed.

$$\cos(\varphi) = \frac{\overline{u_x(t)}}{U} \quad (3a)$$

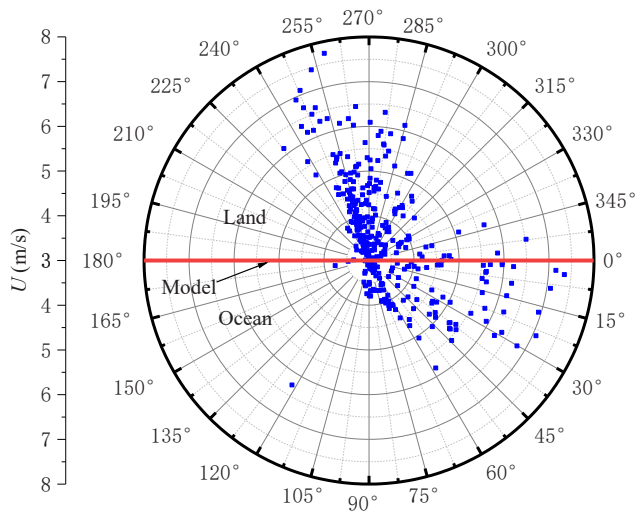


Fig. 5. The U and φ for all 10-minute intervals during the test.

$$\tan(\alpha_m) = \frac{\overline{u_z(t)}}{U} \quad (3b)$$

where φ is the wind direction angle, measured clockwise from 0° along the bridge. α_m is the wind angles of attack, with positive wind direction originating from the main beam.

$$I_i = \frac{\sigma_i}{U} (i = u, v, w) \quad (4)$$

where $\sigma_i (i = u, v, w)$ is the standard deviation of fluctuating wind speed $u(t)$, $v(t)$ and $w(t)$.

$$L_i = \frac{U}{\sigma_i^2} \int_0^\infty R_i(\tau) d\tau \quad (i = u, v, w) \quad (5)$$

where σ_i and R_i represent the standard deviation and autocorrelation function of the different fluctuating components ($u(t)$, $v(t)$ and $w(t)$).

Preliminary analysis indicated that the wind speed characteristics from the three anemometers were consistent, so the data from the wind speed anemometer in the second span of the model were used to represent the wind field characteristics. Fig. 4 shows the U , φ , α_m , I_u and I_v from 08:00:00–14:30:00 on October 17, 2023. Fig. 5 presents U and φ for all 10-minute intervals during the test. Fig. 6 displays the relationship between α_m and U , while Figs. 7 and 8 show the variations of I_u and I_v with U , respectively.

Fig. 9 shows the probability distribution histograms of turbulence length scales of L_u , L_v and L_w . As the primary focus is on the VIV of the model, only the wind field characteristics within the wind speed range ($3 \sim 8$ m/s) capable of inducing VIV are presented in Figs. 5–9.

As shown in Fig. 4, many of the wind speeds recorded on the observation date fall within the VIV lock-in region (4.6 m/s to 7.7 m/s). The φ ranges from 236° to 290° , corresponding to the specific α_m . The test site exhibits low turbulence intensity, indicating that the wind field conditions are well-suited for the experiment. As shown in Figs. 5–8, most wind directions are nearly perpendicular to the model's axial direction, with φ values predominantly ranging between 240° and 285° . The range of φ is likely to induce VIV in the model, leading to significant amplitude responses [13]. As shown in Fig. 9, these turbulence length

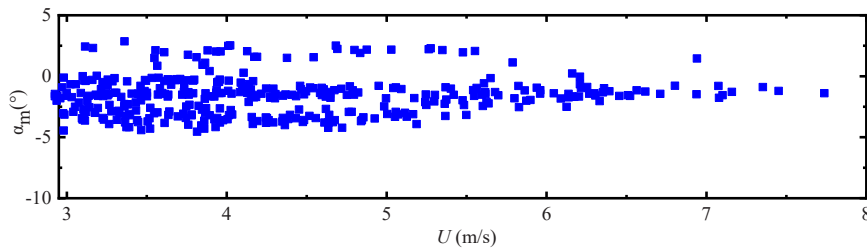


Fig. 6. The α_m varies U .

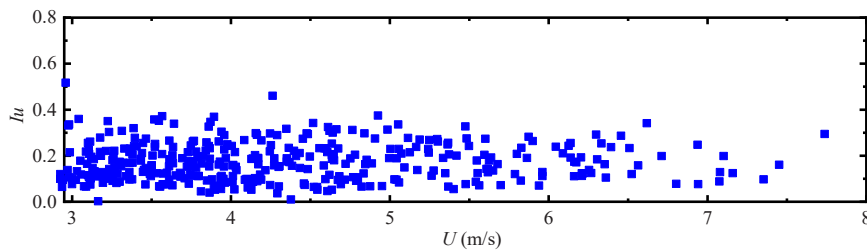


Fig. 7. The I_u varies U .

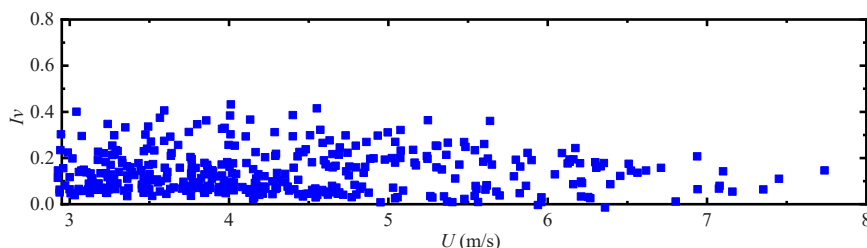


Fig. 8. The I_v varies U .

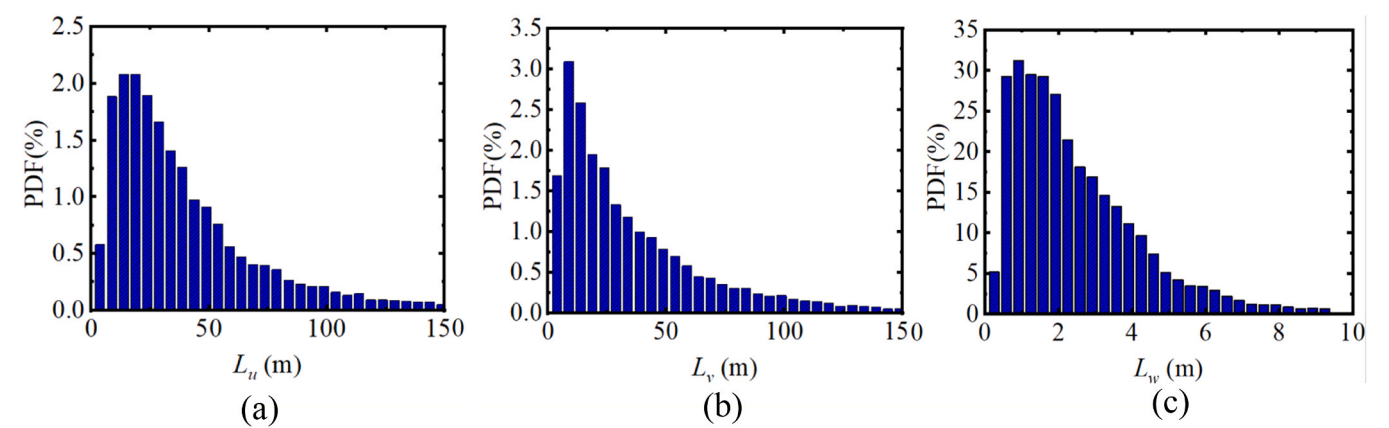


Fig. 9. The probability distribution histograms of turbulence length scales: (a) L_u ; (b) L_v ; (c) L_w .

Table 1
Information summary of VIV events.

No.	Frequency (Hz)	Average amplitude (mm)	Maximum amplitude (mm)	Average U (m/s)	φ (°)	α_m (°)	L_u	Vibration duration (s)
1	2.26	2.6	6.9	4.95	255	−1.8	0.18	65
2	2.3	1.9	6.6	4.82	249	−1.6	0.17	50
3	2.26	2.2	4.93	7.32	258	−2.4	0.23	50
4	2.24	1.6	3.1	6.4	252	−1.6	0.19	40
5	2.23	3.4	12.6	7.3	255	−3.7	0.15	130
6	2.27	2.9	6.6	6.2	258	−2	0.20	75
7	2.22	2.1	5.7	7.1	237	−2.1	0.15	48
8	2.3	1.1	3.5	7.1	230	−2.4	0.20	40
9	2.26	2.0	11.81	7.2	244	−1.6	0.22	81
10	2.3	1.4	4.3	6.6	240	−2.1	0.18	45
11	2.27	1.4	3.6	6.1	223	−2.4	0.20	60
12	2.22	2.3	6.7	6.5	250	−2.2	0.15	65
13	2.23	1.6	4.3	4.1	241	−0.6	0.15	35
14	2.26	1.3	3.3	6	250	−2.1	0.17	87
15	2.26	1.5	3.9	5.3	255	−1.8	0.14	50
16	2.26	2.3	7.0	5.5	267	−1.7	0.12	50
17	2.25	3.5	7.2	7.8	253	−2	0.16	35
18	2.27	2.4	5.2	6.2	256	−1.9	0.21	45
19	2.26	4.9	10.5	7.7	248	−2.8	0.21	42
20	2.31	2.7	5.6	7.4	254	−2.1	0.18	45
21	2.27	2.3	5.6	7.0	258	−1.7	0.21	47
22	2.28	2.8	4.7	7.3	253	−1.5	0.2	25
23	2.27	4.1	7.7	6.3	43	−1.9	0.2	22
24	2.26	2.0	4.4	6.3	237	−1.7	0.19	50
25	2.26	3.1	11.9	7.1	246	−3.8	0.21	85
26	2.25	2.4	6.6	7.4	246	−2.1	0.23	53
27	2.32	2.7	6.2	7.3	267	−2.7	0.21	35
28	2.26	2.4	6.2	6.6	271	−2.86	0.21	55
29	2.28	2.1	4.2	5.6	67	−2.6	0.16	40
30	2.23	1.8	4.5	5.8	263	−2.1	0.22	35
31	2.28	1.7	4.4	6.2	269	−2.3	0.22	40
32	2.22	4.2	7.7	6.8	268	−2.7	0.24	25
33	2.29	2.8	8.5	6.4	288	−4.2	0.24	37
34	2.27	1.2	3.1	6	284	−3.3	0.22	70
35	2.25	3.8	10.0	7.5	272	−3.2	0.2	66
36	2.26	1.9	5.5	5.8	288	−3.1	0.21	110
37	2.24	3.4	6.7	7.1	270	−3.3	0.26	33
38	2.27	1.8	7.2	5.3	271	−2.7	0.15	65
39	2.23	2.3	4.4	5.4	275	−2.8	0.21	40
40	2.23	1.6	4.3	5.5	293	−3	0.18	72
41	2.23	0.9	2.1	4.6	286	−2.7	0.23	90
42	2.23	1.8	3.3	5.5	278	−2.4	0.21	22
44	2.23	4.2	14.0	5.4	282	−5.1	0.21	45
45	2.25	1.5	6.6	7.42	246	−2.17	0.23	53
46	2.25	3.1	8.6	6.2	289	−4.2	0.29	70

Note: The average amplitude represents the mean amplitude during a VIV event. In the VIV test of the full-bridge aeroelastic model of R4:1-CGB under natural wind field, the duration of VIV events is relatively short. If the 10-minute interval wind data were still used to calculate the wind parameters during VIV occurrence, it would introduce deviations in key parameters such as U , φ , and α_m . To ensure the accuracy and reliability of the test data, when analyzing the wind field data corresponding to the VIV characteristics of the model, the 1-minute interval wind data was adopted for parameter extraction and selection.

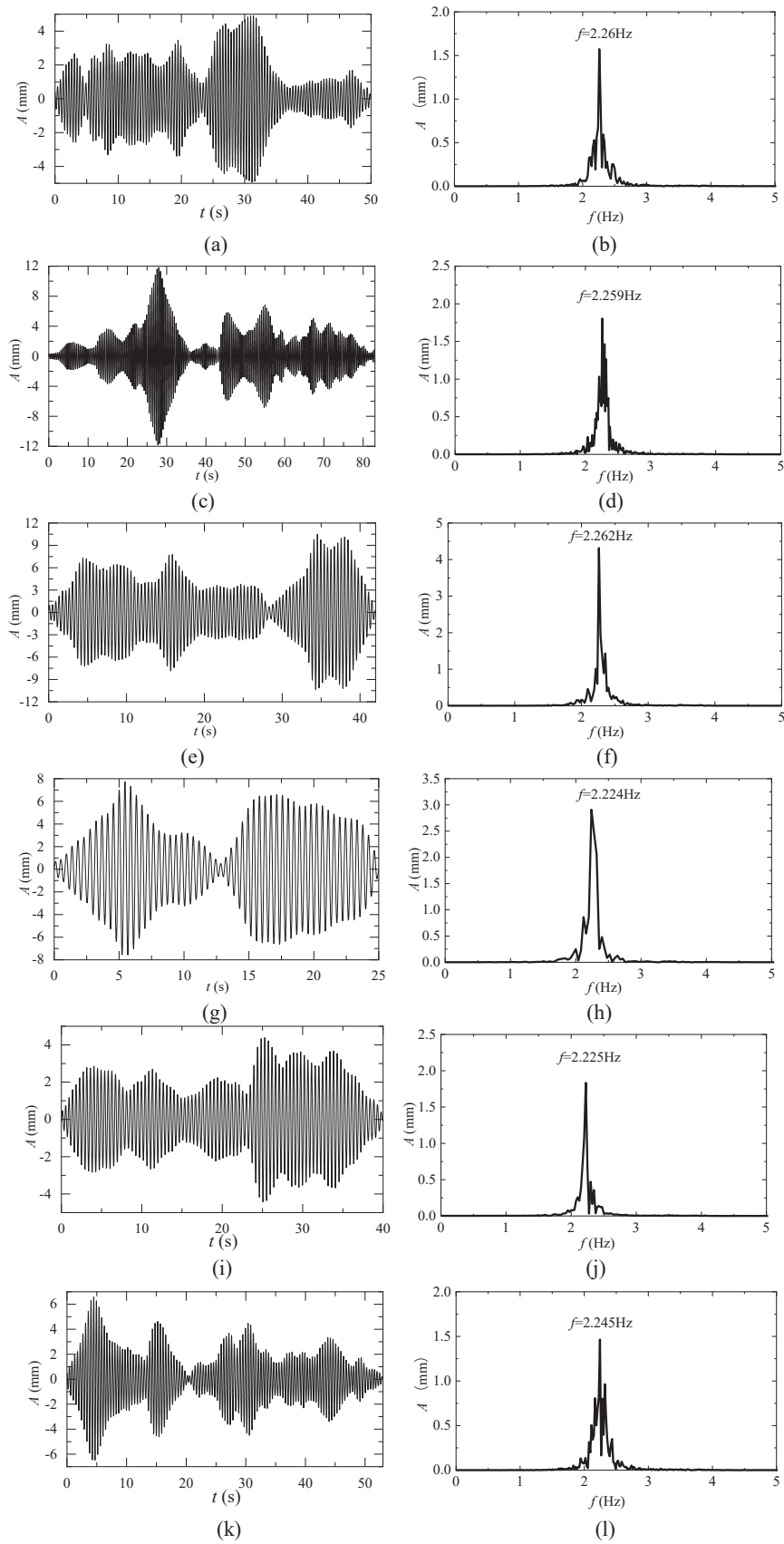


Fig. 10. The displacement time histories and amplitude spectra for the VIV events: (a), (c), (e), (g), (i) and (k) of the time history of No. 3, 9, 19, 32, 39 and 45 VIV events; (b), (d), (f), (h), (j) and (l) of the amplitude spectrum of No. 3, 9, 19, 32, 39 and 45 VIV events.

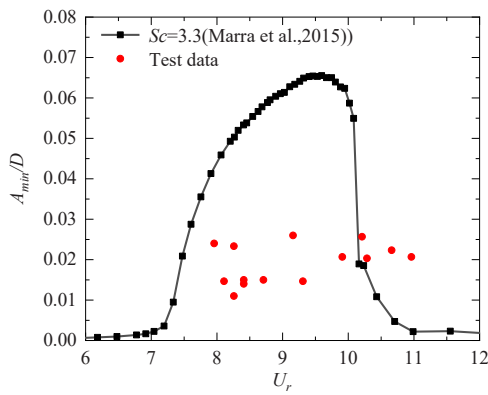


Fig. 11. The VIV responses.

scales follow geometric scaling, thereby rendering the wind field suitable for conducting VIV investigations on the model bridge.

3.2. VIV of the R4:1-CGB

Several VIV events were recorded during the test, and the signal from the No.7 vertical acceleration sensor was analyzed. Table 1 presents detailed information on the selected VIV events. Owing to the rapid fluctuations in the natural wind field, the onset and cessation of significant VIV amplitudes were recorded as distinct VIV events, facilitating a more precise analysis of the relationship between wind parameters and VIV responses.

Based on the data in Table 1, the frequency range of VIV is 2.22 ~ 2.32 Hz, which closely matches the first-order vertical bending frequency of the R4:1-CGB. The ranges for each parameter are as follows: U from 4.1 to 7.8 m/s; φ from 223° to 293° and 43° to 67°; α_m from -5.1° to -0.6°; I_u from 0.12 to 0.29; average amplitude (A_{ave}) from 0.9 to 4.9 mm; and maximum amplitude (A_{max}) from 2.1 to 14 mm.

Fig. 10 displays the displacement time history signal and amplitude of several VIV events. Fig. 11 compares the VIV events with the experimental results reported by Marra et al. [10]. For the selected VIV events, the wind yaw angle is within 10°, the natural wind field's α_m is approximately -3°, and the Sc number of the R4:1-CGB is 3.1, which is close to the value of 3.3 in the experimental results of Marra et al. [10]. The vertical bending response of the R4:1-CGB is characterized using the reduced wind speed, $U_r = U/f_a D$, and the dimensionless displacement, A_{max}/D , where U denotes the average wind speed during the test, f_a represents the model's first-order vertical bending frequency, D is the model height, and A_{max} is the maximum amplitude of the VIV event.

Fig. 10 illustrates that the VIV mode in the natural wind field for the

R4:1-CGB remains predominantly single-mode, with a distinct dominant frequency and a narrow spectrum. The displacement time history indicates that the VIV amplitude fluctuates owing to variations in the natural wind field, and that this amplitude is dependent on wind speed. Fig. 11 demonstrates that most VIV data points fall within the lock-in region reported by Marra et al. [10], with the amplitude increasing as wind speed rises. Despite the small Sc number and the presence of a specific α_m in this study, the VIV amplitude remains significantly smaller than the test results reported by Marra et al. [10]. Some data points fall outside the lock-in region, primarily due to considerable fluctuations in the natural wind field's wind speed. Calculating the average wind speed over a 1-minute interval introduces some deviation; however, most wind speed values in the time history remain within the lock-in region. Also the differences may be ascribed to various experimental uncertainties, e. g., the model fabrication accuracy, dimension, boundaries, installation status, flow conditions (attack angle, yaw angle, turbulence, velocity, etc.), Reynolds number, and apparatus measurement accuracy.

To further clarify the effects of wind characteristics on the VIV responses of the R4:1-CGB, detailed analyses of recorded VIV events were conducted. Fig. 12(a) illustrates the relationship between the A_{max} and U under different β . Fig. 12(b) illustrates the relationship between the A_{max} and U under different α_m . Fig. 12(c) illustrates the relationship between the A_{max} and U under different I_u .

As shown in Fig. 12, the β inducing VIV in the R4:1-CGB ranges from 0° to 50°. It is evident that the amplitude of VIV increases as the wind direction approaches perpendicular to the bridge axis. Thus, in natural wind conditions, wind directions perpendicular to the model axis remain the most critical scenario inducing significant VIV. Additionally, the displacement amplitude variation due to changes in β and I_u is relatively small. Unlike controlled wind tunnel tests, the rapid wind speed fluctuations in natural wind fields make it difficult to maintain stable I_u and α_m during VIV. Nevertheless, observations indicate that, under similar wind speed conditions, VIV amplitudes tend to increase with decreasing I_u and increasing α_m .

VIV response is very sensitive to the section details, so the results for the 4:1 rectangular cylinder may be different with other sections. As is known, the 4:1 rectangular cylinder is often considered a simplified bridge deck since its aerodynamic performance is qualitatively similar to some bluff bridge decks, so the experimental results may provide a benchmark for checking the prediction accuracies of mathematical models and computational dynamic simulations of the VIV responses under skew winds.

4. Conclusions

The VIV characteristics of a rectangular 4:1 cross-section continuous girder bridge were experimentally investigated under natural wind

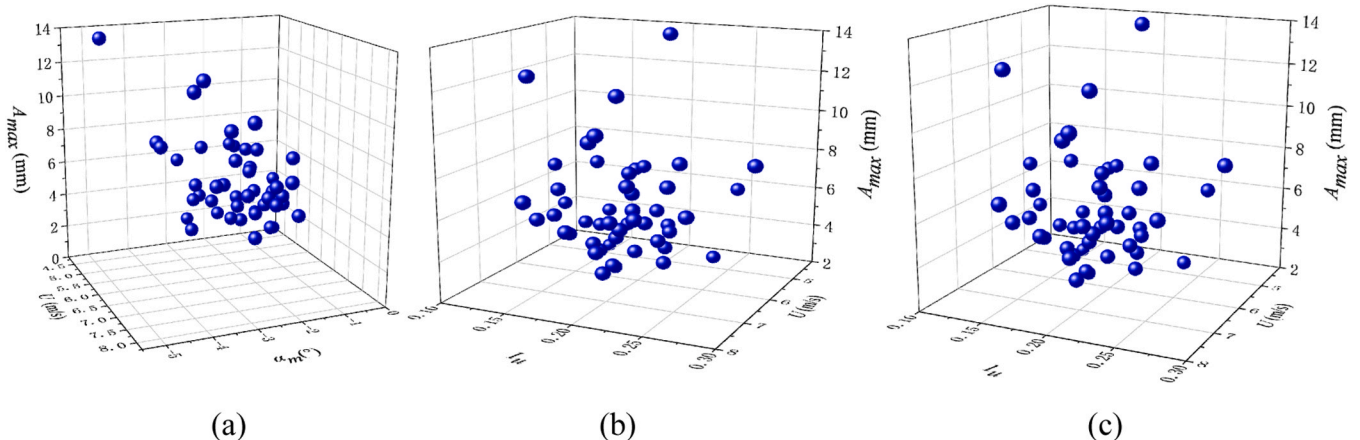


Fig. 12. The A_{max} of VIV: (a) The A_{max} of VIV under different U_r and β ; (b) The A_{max} of VIV under different U_r and α_m ; (c) The A_{max} of VIV under different U_r and I_u .

conditions. The primary conclusions are summarized as follows:

- (1) During the observation period, the U ranged from 0 to 8 m/s, with the φ primarily between 240° and 258° , meeting favorable test conditions. The α_m were mostly negative, showing minimal variation with U . Both I_u and I_v decreased as U increased.
- (2) Several VIV events were captured during testing. The VIV mode in the natural wind field was predominantly single-mode, with a distinct dominant frequency and a narrow frequency spectrum. However, displacement time histories showed amplitude fluctuations caused by the inherent randomness of natural wind conditions, with amplitude variations closely related to wind speed U .
- (3) Most measured VIV data points fall within the lock-in region established by wind tunnel tests, with amplitudes increasing with wind speed. However, observed amplitudes were significantly smaller compared to wind tunnel section model test results. A few data points fell outside the lock-in region, likely due to considerable fluctuations in natural wind speeds. Deviations might also arise from averaging wind speeds over 1-minute intervals.
- (4) The β inducing VIV in the R4:1-CGB ranges from 0° to 50° . The VIV amplitudes increase as wind direction approaches perpendicularity to the bridge axis, indicating that wind perpendicular to the model axis represents the most critical scenario for VIV. Under similar wind speeds, amplitudes increase as I_u decreases and α_m increases.

CRedit authorship contribution statement

Fuyou Xu: Writing – review & editing, Writing – original draft, Validation, Supervision, Methodology, Conceptualization. **Bo Wang:** Writing – review & editing, Writing – original draft, Software, Methodology, Formal analysis, Data curation. **Mingjie Zhang:** Writing – review & editing, Writing – original draft, Methodology.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

The research is jointly supported by the National Science Foundation of China (52125805, 51978130), both are gratefully acknowledged.

Data availability

Some data, models, or code that support the findings of this study are available from the corresponding author upon reasonable request.

References

- [1] Laima S, Li H, Chen W, Ou J. Effects of attachments on aerodynamic characteristics and vortex-induced vibration of twin-box girder. *J Fluids Struct* 2018;77:115–33.
- [2] Larsen A, Larose GL. Dynamic wind effects on suspension and cable-stayed bridges. *J Sound Vibration* 2015;334:2–28.
- [3] Xu F, Ying X, Li Y, Zhang M. Experimental explorations of the torsional vortex-induced vibrations of a bridge deck. *J Bridge Eng* 2016;21:04016093.
- [4] Zhao G, Wang Z, Zhu S, Hao J, Wang J. Experimental study of mitigation of wind-induced vibration in asymmetric cable-stayed bridge using sharp wind fairings. *Appl Sci* 2022;12:242.
- [5] He X, Zhang B, Zou Y, Yi L, Cai C. Experimental research on Wind-resistant performance of mainbridge of wuhu changjiang river rail-Road bridge. *Bridge Constr* 2019;49.
- [6] Li H, Laima SJ, Ou JP, et al. Investigation of vortex-induced vibration of a suspension bridge with two separated steel box girders based on field measurements[J]. *Eng Struct* 2011;33(6):1894–907.
- [7] Wu T, Kareem A. An overview of vortex-induced vibration (VIV) of bridge decks [J]. *Front Struct Civ Eng* 2012;6(4):335–47.
- [8] Ma Zhaoyu, Xu Fuyou, et al. Design, fabrication, and dynamic testing of a large-scale outdoor aeroelastic model of a long-span cable-stayed bridge[J]. *Eng Struct* 2022;256:114012.
- [9] Xu Fuyou, Ma Zhaoyu, et al. A new method studying wind engineering of bridges: Large-scale aeroelastic model test in natural wind[J]. *J Wind Eng Ind Aerodyn* 2020;202.
- [10] Marra AM, Mannini C, Bartoli G. Measurements and improved model of vortex-induced vibration for an elongated rectangular cylinder[J]. *J Wind Eng Ind Aerodyn* 2015;147:358–67.
- [11] Zhang M, Xu F, Øiseth O. Aerodynamic damping models for vortex-induced vibration of a rectangular 4: 1 cylinder: comparison of modeling schemes[J]. *J Wind Eng Ind Aerodyn* 2020;205:104321.
- [12] Zhang M, Wu T, Xu F. Vortex-induced vibration of bridge decks: describing function-based model[J]. *J Wind Eng Ind Aerodyn* 2019;195:104016.
- [13] Wang Bo, Zhang Mingjie, Xu Fuyou. Experimental investigation on the VIV of a rectangular 4:1 cylinder under skew winds[J]. *J Wind Eng Ind Aerodyn* 2022;229.